PASSMOE-P: ENHANCING PASSWORD GUESSING USING LARGE LANGUAGE MODELS WITH PATTERN-SPECIALIZED MIXTURE-OF-EXPERTS

Beilei Zhang[1⋆], *Yuke Ma*[1⋆], *Tao Hu*[1], *Yanzhao Gao*[1], *Hongtao Yu*[1], *Fan Zhang*[5], *Hailong Ma*[1,2,3,4†]

[1] Information Engineering University, Zhengzhou, China
[2] Key Laboratory of Cyberspace Security, Ministry of Education of China, China
[3] National Key Laboratory of Advanced Communication Networks, China
[4] Laboratory of Cyberspace Endogenous, Henan Province, China
[5] Fudan University, Shanghai, China

ABSTRACT

Efficient guessing techniques are crucial for password security auditing. However, monolithic large language models (LLMs) face fundamental scalability bottlenecks due to capacity-conflict problems across heterogeneous password patterns. We propose PassMoE-P, the first Mixture-of-Experts (MoE) framework with pattern-specialized LLMs. Our core innovation integrates domain knowledge of behavioral semantics into three parameter-efficient LoRA experts: a PII Semantic Expert with Bayesian priors for personalized structure modeling, a High-Entropy Expert featuring entropy-adaptive variational sampling, and a Morph Transformation Expert regularized by Levenshtein-distance constraints. A hybrid CNN-GRU gating network enables dynamic routing based on quantified semantic features (PII score, structural entropy, leetspeak score). Evaluated on 70.5 million real-world passwords across heterogeneous datasets, PassMoE-P significantly outperforms state-of-the-art baselines in attack success while reducing VRAM consumption by 50% and increasing throughput 4.8× compared to a monolithic baseline.

***Index Terms*—** Large Language Models, Mixture-of-Experts, Password Guessing, Low-Rank Adaptation, Semantic-Aware Gating.

1. INTRODUCTION

Password security remains a critical challenge in cyberspace, with compromised credentials implicated in 45% of data breaches according to the 2025 Verizon DBIR report [16]. Traditional probabilistic approaches, including Markov models [2] and Probabilistic Context-Free Grammars (PCFG) [6], demonstrate limited efficacy against modern password sets. Password modeling research has evolved from probabilistic methods to modern neural architectures. Traditional approaches capture shallow patterns but fail to model semantic behaviors like PII derivation or phrase transformations, achieving 35% success on targeted subsets [18] [4]. Recent architectural innovations offer pathways to resolve this constraint. Mixture-of-Experts (MoE) frameworks like ST-MoE [21] enable conditional computation through dynamic routing, while syntax-aware gating techniques [12] demonstrate feature-dependent specialization. These advances remain disconnected from password security's unique requirements, where semantic stratification demands specialized routing heuristics beyond token-level features.

The emergence of large language models (LLMs) has transformed password security research: PassGPT pioneered transformer-based password modeling [14], while PassLLM established new benchmarks through low-rank adaptation of Mistral-7B [22]. However, these monolithic architectures had a fundamental limitation across semantic categories when modeling heterogeneous password syntax [13]. This deficiency manifests acutely when reconciling three divergent password creation paradigms: PII-derived patterns (e.g., 'Zhang1998'), high-entropy random strings (e.g., 'xq9!k'), and lexical transformations (e.g., leetspeak 'P@$$w0rd'). While Mixture-of-Experts (MoE) architectures have succeeded in scaling LLMs [15], existing implementations primarily decomposed tasks at the neural layer level such as replacing FFN blocks [11]. A critical limitation of prior work about designing experts with password semantics constrains model interpretability and generalization.

To bridge this gap, we introduce PassMoE-P, the first framework to apply pattern-driven specialization within an MoE architecture specifically for password guessing. Our core innovation lies in decomposing password generation at the semantic-pattern level rather than conventional token-level routing [11]. Specifically, Expert A specializes in PII mapping like name and birthday structure); Expert B masters stochastic character combinations; Expert C focuses

Supported by the National Key Laboratory of Advanced Communication Networks (Grant No. FFX24641X028) and the Science and Technology Major Project of Henan Province, China (Grant No. 241110210200).

⋆ First Author and Second Author contribute equally to this work.
† Corresponding Author: longmanclear@163.com

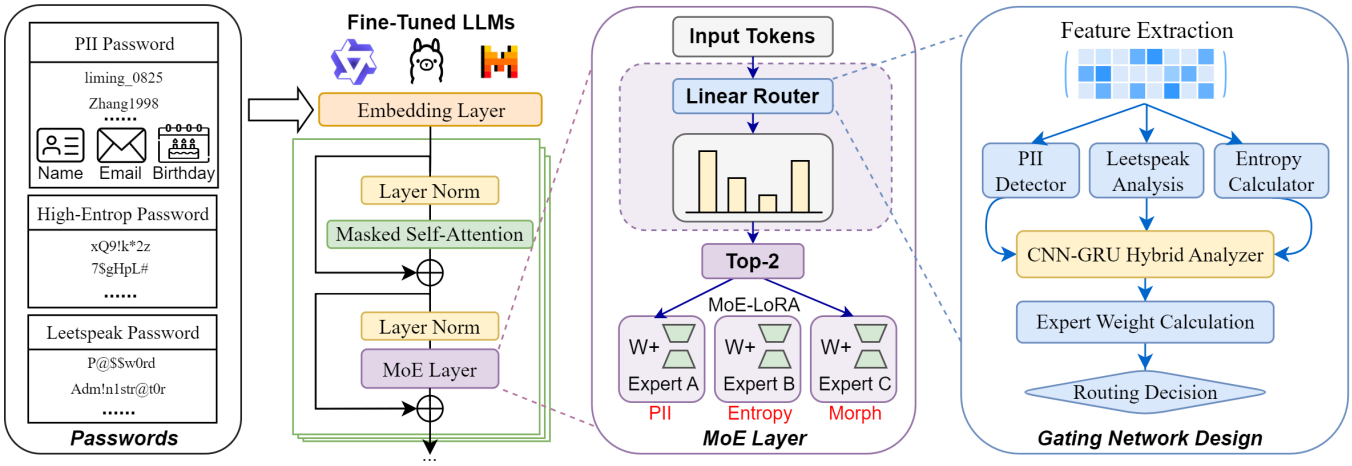


Fig. 1. The architecture of PassMoE.

on phrase transformation rules including leetspeak. Unlike token-level MoE, our gating network dynamically routes tasks based on password semantics—activating Expert A when PII is detected, or Expert C when leetspeak fragments emerge—enabling real-time pattern adaptation [9].

In summary, our main contributions are as follows:

- We propose PassMoE-P, a novel pattern-based MoE framework that decomposes password generation at the semantic level to design experts for PII-derived, high-entropy, and phrase-transformed paradigms.
- We introduce a gating mechanism that employs a top-2 routing strategy. This allows the framework to dynamically combine the strengths of multiple experts to effectively handle complex, mixed-pattern passwords.
- We comprehensively evaluate our approach on large-scale real-world password datasets, providing detailed analysis of guessing success, computational efficiency, and routing decision interpretability.

2. METHODOLOGY

PassMoE-P combines a shared LLM backbone, a gating network, and three LoRA-based experts [7]. Inputs (PII or prefixes) are encoded by the shared LLM, with the gating network routing outputs to specialized experts. This leverages MoE's efficiency: only relevant experts activate per input, reducing computation [1].

2.1. Backbone LLM Selection and Fine-Tuning

We evaluate multiple state-of-the-art large language models as foundational backbones for PassMoE-P, including several models from the Qwen2.5 series (0.5B, 3B, and 7B) [3], Llama3-8B [10], and Mistral-7B [8], to analyze the impact of model scale on performance. All models undergo task-adaptive pretraining on the COMB dataset using: a. Masked token prediction (15% masking rate); b. Causal language modeling; c. Pattern-specific contrastive learning.

2.2. Pattern-Driven Expert Design

Recognizing that user password creation follows discernible behavioral paradigms, we design three specialized experts, each targeting a specific pattern. All experts are built upon the shared backbone using Low-Rank Adaptation (LoRA) with a rank of (LoRA) with a rank of $r = 32$. This approach reduces the number of trainable parameters by 98.6% compared to full fine-tuning of the entire model for each expert, making the MoE framework remarkably parameter-efficient.

2.2.1. PII Semantic Expert (Expert A)

The PII Semantic Expert (Expert A) is designed to model passwords derived from personally identifiable information. It features a Bidirectional LSTM (BiLSTM) layer atop its LoRA adapters, which processes structured PII tuples like [Name, BirthYear] and injects their semantic context into the attention mechanism. To further enhance generation accuracy, a Bayesian prior layer is implemented to model the probabilistic relationship between PII types and common suffixes.

2.2.2. High-Entropy Expert (Expert B)

The High-Entropy Expert (Expert B) specializes in modeling stochastic character combinations. Unlike standard random generators, we employ an LLM to capture the "human randomness" bias—cognitive patterns where users attempt to type randomly but inadvertently follow keyboard

layouts (e.g., QWERTY walks) or predictable shifts. The generation is controlled by entropy-adaptive sampling: the probability distribution for the next token is sampled from $\mathcal{N}(\mu, \sigma)$, where the variance is scaled by the Shannon entropy $H(\text{prefix})$ of the preceding sequence ($\sigma = 1.5 \times H(\text{prefix})$).

This scaling factor creates a dynamic *exploration-exploitation* mechanism. For low-entropy prefixes (e.g., "1234"), σ approaches zero, forcing the expert to converge deterministically on highly probable completions. Conversely, as $H(\text{prefix})$ increases (signaling a high-entropy intent), σ scales up, encouraging the model to explore diverse continuations. This prevents the "mode collapse" often seen in standard LLMs when generating random strings.

2.2.3. *Morph Transformation Expert (Expert C)*

The Morph Transformation Expert (Expert C) is engineered to master pattern-based transformations, most notably Leetspeak (e.g., `a->@`, `s->$`) and common string manipulations found in password reuse. Its architecture is augmented by incorporating a Levenshtein distance-based regularization term into the loss function, defined as $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot \text{LevSim}(pw_{\text{gen}}, \mathcal{D}_{\text{leet}})$. This regularization encourages the expert to generate minor morphological variations of an input seed by measuring the similarity LevSim of its output to a dictionary of common Leetspeak transformations($\mathcal{D}_{\text{leet}}$).

2.3. Semantic-Aware Gating Network

Instead of simplistic token-level routing, we use a semantic-aware gating network, $\mathcal{G}(\cdot)$, a hybrid CNN-GRU model that extracts semantic features (PII score, leetspeak score, structural entropy) from inputs and projects them through a softmax layer to generate expert routing weights α_k. We employ a top-2 routing strategy, activating the two experts with the highest weights for each input. **Efficiency and Robustness:** While the gating network introduces a feature extraction step, its computational overhead is negligible compared to the backbone LLM. The gating inference occurs only once per password prefix, whereas the LLM decoding runs iteratively for every token. Furthermore, the top-2 strategy ensures robustness against feature extraction errors (e.g., missed PII detection in non-Western names); if the primary semantic feature is ambiguous, the secondary expert is still activated to maintain generation diversity.

2.4. MoE-Enhanced Password Generation Algorithm

We propose an adaptive algorithm that leverages the semantic-aware gating network to dynamically route all input types to the most appropriate expert. Algorithm 1 processes semantic inputs (either password prefixes or structured auxiliary data) through a single pipeline. The gating network extracts discriminative features (PII score, Leetspeak score, structural entropy) to select the optimal experts, which then generates candidates via expert-specific strategies (e.g., PII semantic integration, entropy-adaptive sampling). Candidate processing follows beam search with probability thresholding [20].

Algorithm 1: MoE-Enhanced Password Generation

Input: **Experts:** $\{E_A, E_B, E_C\}$, **Gating Network:** $\mathcal{G}$, **Semantic Input:** s (prefix or auxiliary data), **Max Passwords:** N, **Prob Threshold:** τ, **Beam Width:** b *// New parameter for candidate control*
Output: **Password Set:** C

Initialize: $C \leftarrow \emptyset$ *// Completed password collection*
$Q \leftarrow [s]$ *// Queue for active prefixes*
While
  $Q \neq \emptyset$ **and** $|C| < N$
$s_{\text{cur}} \leftarrow Q.\text{pop}(0)$ *// Dequeue current prefix*
// Feature Extraction
$f_{\text{pii}} \leftarrow \frac{\#\text{NER}_{PII}}{|s_{\text{cur}}|}$ *// PII detection score*
$f_{\text{leet}} \leftarrow \frac{\#\text{matched rules}}{|s_{\text{cur}}|}$ *// Leetspeak score*
$H \leftarrow -\sum p(c) \log p(c)$ *// Structural entropy*
// Expert Routing (Top-2)
$[\alpha_A, \alpha_B, \alpha_C] \leftarrow \mathcal{G}(s_{\text{cur}})$ *// Get gating weights*
$\{e_1^*, e_2^*\} \leftarrow \underset{k \in \{A,B,C\}}{\text{top2}} (\alpha_k)$ *// Select top-2 experts*
// Parallel Expert Generation
candidates $\leftarrow \emptyset$
foreach $e^* \in \{e_1^*, e_2^*\}$ **do**
  partial_cands $\leftarrow e^*.\text{beam_search}(s_{\text{cur}}, \text{beam_size} = b/2)$
  candidates $\leftarrow$ candidates $\cup$ partial_cands *// Merge candidate sets*
end
// Candidate Processing
foreach $c \in$ *candidates* **do**
  if c *ends with* <EOS> ***and*** $\Pr(c) > \tau$ **then**
    $C \leftarrow C \cup \{c\}$ *// Add complete password*
  else if $\Pr(c) > \tau/10$ **then**
    $Q.\text{append}(c)$ *// Requeue prefix for expansion*
end
return C

3. EXPERIMENTS AND RESULTS

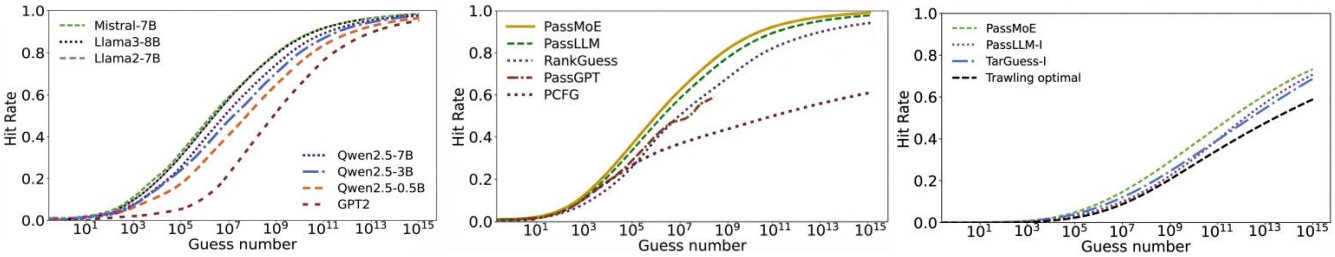
3.1. Experimental Setup

3.1.1. *Datasets*

We evaluated PassMoE-P using five complementary datasets representing distinct password creation behaviors, including CSDN, Dodonew, Rockyou, 000Webhost and LinkedIn. Feature extraction uses these real-world pattern databases. These datasets collectively form the 70.5M password COMB corpus (see Table 1) used for task-adaptive pretraining.

3.1.2. *Baselines and Metrics*

We compared against five state-of-the-art models: PassLLM [22], PassGPT [14], TarGuess [17], RankGuess [19] and



(a) The impact of base models (b) Trawling password model (c) PII-based targeted guessing scenarios

Fig. 2. The comparison of password guessing performance.

PCFGv4 [5]. Evaluation metrics included *Success Rate@1B* (passwords cracked within 1 billion guesses), *Guess Reduction*, and *Pattern Recall* (expert-specific accuracy).

Table 1. COMB Dataset Composition

Dataset	Passwords	Language	Characteristics
CSDN	6.4M	Chinese	Programmer community, mixed entropy
Dodonew	16.2M	Chinese	E-commerce, PII-heavy patterns
Rockyou	32.6M	English	Social forum, Leetspeak variants
000Webhost	15.3M	English	Web hosting, random strings
LinkedIn	6.5M	English	Professional hybrid patterns
Total	70.5M		Comprehensive behavioral coverage

3.2. Password Guessing Performance

As illustrated in Fig. 2, PassMoE-P demonstrates well performance across various password guessing scenarios. In the general trawling attack evaluation (Fig. 2b), the framework's hit rate consistently surpasses all baselines, including contemporary approaches such as PassLLM [22] and PassGPT [14], and it is significantly more effective than traditional probabilistic methods like PCFG. This performance advantage is particularly pronounced in targeted guessing scenarios that leverage Personally Identifiable Information (PII) (Fig. 2c), where PassMoE-P again establishes a new benchmark. It achieves a hit rate of approximately 15% within a thousand guesses, exceeding the performance of specialized models like PassLLM-I and TarGuess-I. The efficacy of the framework is fundamentally supported by its backbone model selection (Fig. 2a), where the evaluation confirms that larger-scale models like Mistral-7B and Llama3-8B yield significantly higher performance ceilings compared to smaller variants such as the Qwen2.5-3B, Qwen2.5-0.5B, and the legacy GPT-2 architecture.

3.3. Ablation Studies

As shown in Table 2, removing the gating network caused an 18.2% performance drop. In this "w/o Gating" configuration, we replaced the semantic router with a random Top-2 selection strategy, assigning inputs to arbitrary experts. The significant drop confirms that performance gains stem from precise semantic alignment rather than merely increasing model capacity. Disabling expert-specific components (Bayesian priors, entropy sampling, Levenshtein loss) reduced success rates by 5.7-13.6%. Gradient isolation prevented a 3.5% degradation by maintaining routing integrity.

Table 2. Ablation Impact on Combined Success Rate

Configuration	Success Rate	Δ vs Full
Full PassMoE-P	62.7%	-
w/o Gating Network	51.3%	-18.2%
w/o Bayesian Prior (A)	59.1%	-5.7%
w/o Entropy Sampling (B)	54.2%	-13.6%
w/o Levenshtein Loss (C)	58.4%	-6.8%
w/o Gradient Isolation	60.5%	-3.5%

4. DISCUSSION AND CONCLUSION

In this paper, we proposed PassMoE-P, the first pattern-specialized MoE framework for password generation, resolves fundamental scalability bottlenecks in monolithic large language models (LLMs) by integrating domain-driven semantic decomposition. This work demonstrates that decomposing password guessing into specialized behavioral paradigms through a Mixture-of-Experts (MoE) architecture significantly advances both efficacy and efficiency. This specialization approach outperforms monolithic architectures while maintaining computational efficiency through parameter optimization strategies. The integration of pattern-driven expert design with deterministic routing rules creates a robust framework for both trawling and targeted scenarios. Future work will improve the ability of the gating network to model blended patterns through cross-expert attention mechanisms. Additionally, expanding cross-linguistic coverage to cultural PII conventions beyond English-centric datasets will broaden real-world applicability.

Acknowledgement

This work is supported by the National Key Laboratory of Advanced Communication Networks (Grant No. FFX24641X028) and the Science and Technology Major Project of Henan Province, China (Grant No. 241110210200).

Ethical Considerations

This research strictly adheres to the principle of "offensive research for defensive readiness". We utilized historical, publicly available breach datasets (RockYou, LinkedIn) solely to simulate realistic adversarial scenarios. To mitigate privacy risks, all Personally Identifiable Information (PII) was rigorously sanitized prior to training, ensuring the model learns semantic structures (e.g., "Name+Date") rather than specific user credentials. Our goal is to level the playing field by revealing vulnerabilities in modern password patterns, thereby enabling defenders to design more robust strength meters and proactive blocklists.

References

[1] W. Cai, J. Jiang, F. Wang, J. Tang, S. Kim, and J. Huang. A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*, 2025.

[2] C. Castelluccia, M. Dürmuth, and D. Perito. Adaptive password-strength meters from markov models. In *NDSS*, 2012.

[3] A. Cloud. Qwen. `https://www.alibabacloud.com/help/en/model-studio/what-is-qwen-llm`, 2024. Accessed: 2025-07-21.

[4] M. Dürmuth, F. Angelstorf, C. Castelluccia, D. Perito, and A. Chaabane. Omen: Faster password guessing using an ordered markov enumerator. In *International Symposium on Engineering Secure Software and Systems*, pages 119–132. Springer, 2015.

[5] D. He, Z. Liu, S. Zhu, S. Chan, and M. Guizani. Special characters usage and its effect on password security. *IEEE Internet of Things Journal*, 11(11):19440–19453, 2024.

[6] S. Houshmand, S. Aggarwal, and R. Flood. Next gen pcfg password cracking. *IEEE Transactions on Information Forensics and Security*, 10(8):1776–1791, 2015.

[7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.

[8] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. L. Scao, T. Lavril, T. Wang, T. Lacroix, and W. E. Sayed. Mistral 7b, 2023.

[9] Z. Li, W. Han, and W. Xu. A {Large-Scale} empirical analysis of chinese web passwords. In *23rd USENIX Security Symposium (USENIX Security 14)*, pages 559–574, 2014.

[10] Meta. Llama-3-8b. `https://huggingface.co/meta-llama/Meta-Llama-3-8B`, 2024. Accessed: 2025-07-21.

[11] C. Min, W. Wang, Y. Liu, W. Ye, E. Sangineto, Q. Wang, and Y. Zhao. Guiding the experts: Semantic priors for efficient and focused moe routing. *arXiv preprint arXiv:2505.18586*, 2025.

[12] T. Nguyen, N. N. Tran, K. Nguyen, and R. G. Baraniuk. Improving routing in sparse mixture of experts with graph of tokens, 2025.

[13] B. Pal, T. Daniel, R. Chatterjee, and T. Ristenpart. Beyond credential stuffing: Password similarity models using neural networks. In *2019 IEEE Symposium on Security and Privacy (SP)*, pages 417–434. IEEE, 2019.

[14] J. Rando, F. Pérez-Cruz, and B. Hitaj. Passgpt: Password modeling and (guided) generation with large language models. In *ESORICS (4)*, pages 164–183, 2023.

[15] W. Sun, J. Hu, Y. Zhou, J. Du, D. Lan, K. Wang, T. Zhu, X. Qu, Y. Zhang, X. Mo, et al. Speed always wins: A survey on efficient architectures for large language models. *arXiv preprint arXiv:2508.09834*, 2025.

[16] Verizon Business Group. 2025 data breach investigations report. `https://www.verizon.com/business/resources/reports/dbir/`, 2025. Accessed: 2025-03-15.

[17] D. Wang, Z. Zhang, P. Wang, J. Yan, and X. Huang. Targeted online password guessing: An underestimated threat. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, pages 1242–1254, 2016.

[18] M. Weir, S. Aggarwal, B. De Medeiros, and B. Glodek. Password cracking using probabilistic context-free grammars. In *2009 30th IEEE symposium on security and privacy*, pages 391–405. IEEE, 2009.

[19] T. Yang and D. Wang. Rankguess: Password guessing using adversarial ranking. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 682–700. IEEE, 2025.

[20] D. Yu. Adaptive token boundaries: Integrating human chunking mechanisms into multimodal llms. *arXiv preprint arXiv:2505.04637*, 2025.

[21] B. Zoph, I. Bello, S. Kumar, N. Du, Y. Huang, J. Dean, N. Shazeer, and W. Fedus. St-moe: Designing stable and transferable sparse expert models. *arXiv preprint arXiv:2202.08906*, 2022.

[22] Y. Zou, M. An, and D. Wang. Password guessing using large language models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 7799–7818, 2025.